



Xenobots built using frog cells

Scientists have repurposed live frog cells to assemble xenobots, a living, programmable organism. <https://digit.in/feb20-23>



Ancient digestive tract found

A 550-million-year-old fossilised digestive tract was found in a desert in Nevada recently. <https://digit.in/feb20-24>

eton that assists with the movements of limbs. Potentially, the use of spinal implants in tandem with brain implants can further increase the efficacy of the assistive technologies.

When a person decides to take a step, the motor cortex of the brain relays electrical signals down the backbone, where they reach the motor neurons at the lower spinal cord, near the lumbar region. These neurons extend and contract the leg muscles, so that the individual can take that step. If there is an injury along the spinal cord, then the connection between the motor cortex located in the brain and the motor neurons for the leg muscles located in the lower back is interrupted. The motor cortex and the motor neurons themselves are perfectly functional, the paralysis is caused by their inability to communicate with each other.

In 2016, an international team of researchers experimenting on rhesus macaques, managed to restore motor control on paralysed legs. The two macaques used in the experiment had lesions on their spines spanning the thoracic region, which essentially covered half the spinal cords. The condition is temporary for the affected macaques, and they naturally make a full recovery in about a month. However, such lesions accurately simulate injuries in the spinal cord that cause paralysis in humans, and were useful for the

purpose of the study. Now what the researchers did was first use medical probing techniques to identify the neural hotspots in the spine responsible for locomotor control of the leg muscles. This gave them a clue about which regions to simulate with electrical impulses. A brain implant component was plugged into the motor cortex of the macaques, which uses an electrode array to measure brain activity. The design of the array was based on the BrainGate design, an effort by several institutions to develop an implant that allows paralysed individuals to control robotic arms with their thoughts. This brain implant relayed signals from the motor cortex of a monkey to a computer, where they were processed and then relayed back to another implant in the lower spine. This second implant in the spine provided electric stimulations to the motor neurons, responsible for the movement of the legs. Both the monkeys were able to walk because of the combination of implants. Now, the important part here is the use of the computer and wireless transmission of the signals, because using a wired setup along with the required computing hardware would severely restrict the range of motion available to the monkeys. The animals were able to spontaneously start moving when the system was turned on.

Before such a system can be

translated to human patients, there are some kinks that remain to be worked out. The researchers were unable to determine the amount of pressure that the monkeys were able to put on the affected limb. This means that the study could not effectively judge if the locomotion capabilities were entirely restored for the monkeys, or that the legs were still weaker than unaffected monkeys.

Additionally, the implants could not transmit sensory information from the leg back to the brain. This is just one of the many studies conducted in the area. Scientists have tapped into information from the cervical spinal cord to restore precise gripping capabilities in monkeys, successfully stimulated forelimb locomotion using electric signals in sedated mon-

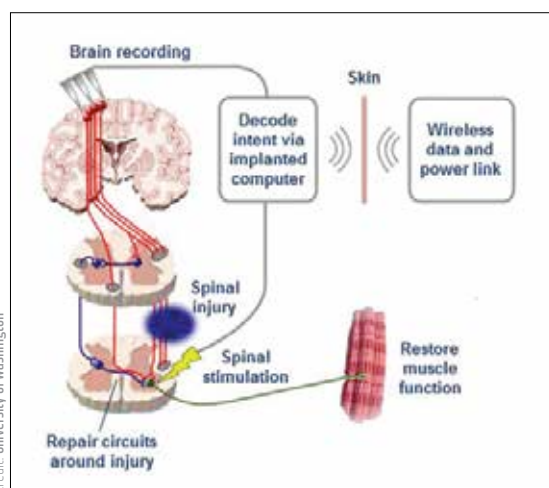


Credit: Brown University

An electrode array like this one could one day be used to restore motor functions in people with spinal cord injuries

keys, pelvic implants in mice have been used to restore locomotion, and electrode implants to restore motor functions in cats have tested to be stable for over six months.

In October 2019, The US Defense Advanced Research Projects Agency (DARPA) gave a \$6.3 million grant to an international team led by Brown University researchers to develop and test an intelligent spine interface that can restore motor functions as well as bladder control for individuals affected by spinal cord injuries. The researchers also hope to use the same implant to restore the sensory functions of the affected regions, and transmit information from the neurons below the site of injury, back to the brain. One of the goals of the project is to use AI to decipher the signals recorded from the spine. Eventually, the team hope to do away with the external computer altogether, and develop a spinal implant with an integrated computer for restoring motor and sensory functions. 



Credit: University of Washington

A schematic for a implant that bypasses a spinal cord injury